

Learning what to inspect: Task-driven multimodal C-scan acquisition for compression-after-impact assessment

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Abstract

Assessing compression-after-impact strength from partial C-scan observations requires allocating measurements that support the evolving strength estimate. We organize this acquisition process around prediction quality throughout a fixed budget, rather than only after acquisition ends. A frozen vision-language model supplies surface-region priors, and a state-dependent actor combines those priors with acquired internal descriptors and the current prediction to select subsequent cells. A separate frozen predictor maps the visible evidence to strength, while policy learning minimizes normalized trajectory error plus terminal error. Nine acquisition strategies are compared through offline image replay on 50 validation specimens from six domains, with checkpoints selected on the same validation set. Under a 25% native-pixel acquisition cap, the main policy achieved 44.286 MPa mean absolute error, compared with 46.910 MPa for geometry-spread. Its trajectory error area was 45.110 MPa, compared with 46.501 MPa for the surface-conditioned open-loop policy. Signed timing contributions place the largest positive main-versus-open-loop difference in the second acquisition-completion stage, after initial internal observations become available. Later negative differences and mixed component comparisons qualify this pattern. The framework connects state-dependent observation allocation to CAI assessment and shows how acquired subsets and their availability over time shape quality-acquisition trade-offs within the evaluated replay setting.

Keywords: compression after impact; task-driven acquisition; C-scan; multimodal inspection; sequential decision making

1 Introduction

Estimating the remaining compressive strength of an impacted composite requires connecting observable damage to a mechanical assessment. Surface appearances, internal damage images and compression-after-impact (CAI) strength measurements provide complementary information about that problem. Public composite-impact datasets associate these observations with corresponding specimens [1, 2]. They support image-based strength assessment and a further question: which internal observations should be acquired when only part of the image will be available? The value of an observation then depends on its contribution to an evolving strength estimate.

C-scan image regression establishes a route from internal damage information to mechanical assessment. Recent work directly predicts CAI strength and impact energy from C-scan images using a convolutional model [3]. Selective acquisition adds a decision before that mapping can be applied to complete input. A policy must choose which missing region to observe next using evidence already available. Two sequences can eventually acquire the same subset yet provide different intermediate predictions because useful information becomes available at different stages. Assessing only the final estimate would overlook this distinction between observation content and availability.

Adaptive ultrasonic inspection provides a precedent for choosing subsequent measurements from earlier observations [4]. Dynamic feature-acquisition methods more generally connect information

requests with prediction utility and acquisition cost [5–7]. For CAI assessment, the relevant utility is how the acquired evidence changes a strength estimate. Visually conspicuous damage may guide an initial observation, but its predictive value must be assessed together with other available information. This motivates a sequential objective that accounts for prediction error throughout acquisition as well as at the endpoint.

The modalities supply different information at different stages. A surface image is available initially, whereas internal content enters the state only after its region has been acquired. Surface cues can therefore initialize a search, and subsequent internal observations can change both the predicted strength and the next acquisition decision. Implementing this progression requires an explicit state interface. The policy must distinguish observed from unobserved regions and relate each permitted descriptor to its location, acquisition history and remaining budget. This interface connects evidence availability to the task objective rather than treating the next region as a fixed function of surface appearance alone.

We develop a state-dependent multimodal acquisition method with three distinct roles. A frozen vision-language model (VLM) supplies surface-region priors, a learned actor selects each internal cell, and a separate frozen predictor estimates CAI strength from the current observation set. The actor is trained using normalized trajectory error area together with terminal absolute error. At each acquisition step, the new internal descriptor and updated prediction enter its state, while the model parameters remain fixed. A common predictor across strategies holds the assessment mapping constant, enabling comparison of the subsets supplied and the timing of their availability.

The study makes three contributions. First, it formulates sequential C-scan acquisition around CAI prediction quality over the acquisition process, rather than only complete-image regression. Second, it connects surface priors, observed internal descriptors and the current strength estimate to subsequent acquisition through a state-dependent policy. Third, it characterizes this policy under a common predictor through same-cap quality, signed timing contributions and matched-quality acquisition requirements. Offline replay on 50 validation specimens compares shared, surface-conditioned and feedback-driven orders. The analysis asks where the resulting prediction value is accumulated and how that process relates to the cost of attaining an empirical quality target; the selection and evaluation scope is specified in Section 4.

2 Related work

2.1 Image-based residual-strength assessment

Image-based assessment connects observable damage with a mechanical quantity that matters after impact. The public datasets of Hasebe and colleagues provide complementary resources: post-impact surface and internal images, and separately recorded compression-after-impact measurements [1, 2]. These resources support specimen-level correspondence between imaging and strength while preserving the difference between a damage observation and a load-bearing property. That distinction motivates the use of CAI error as the acquisition objective here. A region that is visually conspicuous is a candidate for inspection, but conspicuity alone does not establish its value for the strength estimate.

A direct journal neighbour is Mack et al.’s ResNet18-based framework for predicting impact energy and CAI strength from C-scan damage images [3]. Their framework combines direct image interpretation with an analysis of feature-importance maps. This establishes image-to-strength regression as an existing research direction. The distinction in the present study is the sequential allocation of partial observations and the associated quality–acquisition trajectory. We do not compare their reported R^2 numerically with ours because the data preparation and evaluation protocols differ.

Image regression and selective observation address related but separable questions. A predictor must interpret whatever information is supplied, while an acquisition policy must decide which

missing information to request. Improving either component can improve the combined system, but changing both together complicates interpretation of a comparison. The common frozen predictor used here deliberately fixes the assessment mapping across acquisition strategies. This design makes the supplied subsets and their ordering the experimental focus, rather than claiming that the selected regressor is the strongest possible model for complete C-scan images.

2.2 Adaptive acquisition for nondestructive inspection

Autonomous ultrasonic acquisition already has a substantial conceptual precedent. Fuentes et al. formulate the selection of inspection locations through Bayesian optimization and robust outlier analysis [4]. Their method sequentially updates a two-dimensional field of novelty indices to select locations with evidence of damage and estimate component damage probability. The pertinent difference is the task endpoint: we optimize a CAI regression trajectory under an image-pixel budget rather than the reported damage-indication objective.

This distinction also clarifies the status of the fixed controls. A geometry-spread order measures broad spatial coverage, while center-first concentrates observations near the image centre. Neither reproduces an entire autonomous inspection system, and serpentine image order alone does not model robotic travel efficiency. The controls instead provide transparent alternatives under identical replay rules. The controls isolate acquisition ordering under a common strength predictor.

2.3 Task-driven sequential information acquisition

Dynamic feature acquisition provides the broader learning formulation. Shim et al. jointly learn a classifier and an acquisition agent, using a set representation of observed features and actions that either acquire information or stop and predict [5]. Janisch et al. formulate costly-feature classification as sequential decision making, with feature requests and classification actions [6]. Their formulations show that prediction quality and information cost can be optimized together. The present policy instead uses a fixed acquisition cap and a frozen regression model, without a learned classification or stopping action. This is an application-specific choice that preserves a common assessment model for the comparison, rather than a general argument against joint learning.

Covert et al. develop a greedy dynamic-selection method based on conditional mutual information and amortized optimization [7]. Their formulation uses a fixed feature-count budget with uniform costs; its regression result relates squared-error optimality to conditional variance. Our whole-cell native-pixel costs and absolute-error trajectory objective differ from those assumptions. We therefore use this work to position task-relevant acquisition, not as an optimality guarantee or a reproduced baseline. Policy-gradient learning supplies the optimization mechanism in our implementation [8], while the manuscript contribution lies in the inspection-state contract, common-predictor comparison and process-level evidence.

Together, these lines locate the work between image-based mechanical assessment and sequential information acquisition. The method combines a surface-derived starting prior with evidence-dependent internal acquisition, then evaluates both intermediate quality and the final strength estimate. This connection is useful precisely because damage indication, classification utility and CAI regression error need not assign the same value to an observation. The supplementary closest-work matrix compares the task endpoints, available information and cost formulations.

3 Task-driven multimodal inspection framework

3.1 Problem formulation

The inspection task is to acquire a limited subset of an internal C-scan image while maintaining an estimate of compression-after-impact (CAI) strength. Let I^S and I^X denote the raw surface and internal images, and let y denote measured CAI strength in MPa. Each image is partitioned into an 8×8 grid indexed by $\mathcal{I} = \{0, \dots, 63\}$. Independent cell encoding produces surface descriptors $S \in \mathbb{R}^{64 \times 512}$ and internal descriptors $X \in \mathbb{R}^{64 \times 512}$. An action reveals one previously unobserved internal descriptor. The raw images define the regions and native-pixel costs; S and X denote features throughout the decision model. All surface descriptors are available before internal acquisition begins.

After t actions, the observed set is Ω_t and its binary indicator is M_t . The cumulative acquisition fraction c_t is the number of unique acquired native image pixels divided by the number of pixels in the complete internal image. Grid boundaries are rounded on the native image, so cell costs need not be identical. We retain these actual pixel counts when checking affordability and evaluating a trajectory. With $B=0.25$, the legal action set contains only unobserved cells whose complete acquisition keeps c_t within B . Acquisition ends when this set is empty. This rule defines a budget-limited experiment without a learned stopping decision.

Let $X_{\text{obs},t} = M_t \odot X$ denote the logical visible buffer, with the cell mask broadcast across descriptor dimensions. A frozen predictor produces $\hat{y}_t = P_{\text{all}}(S, X_{\text{obs},t}, M_t, c_t)$. The policy selects the next cell from the currently legal set using surface information and, for feedback policies, the acquired internal evidence and current prediction. The true strength y is available to training and retrospective scoring, but is excluded from the decision state. This separation allows the acquisition policy to be trained for a downstream assessment objective without supplying the answer when selecting an observation. The comparison then asks which policy supplies a useful sequence of observations to the same predictor.

We evaluate both the trajectory and its final estimate. Let $e_t = |\hat{y}_t - y|$ and let $e(c)$ hold the most recently available error constant until the next acquisition completes. The normalized trajectory error and objective are

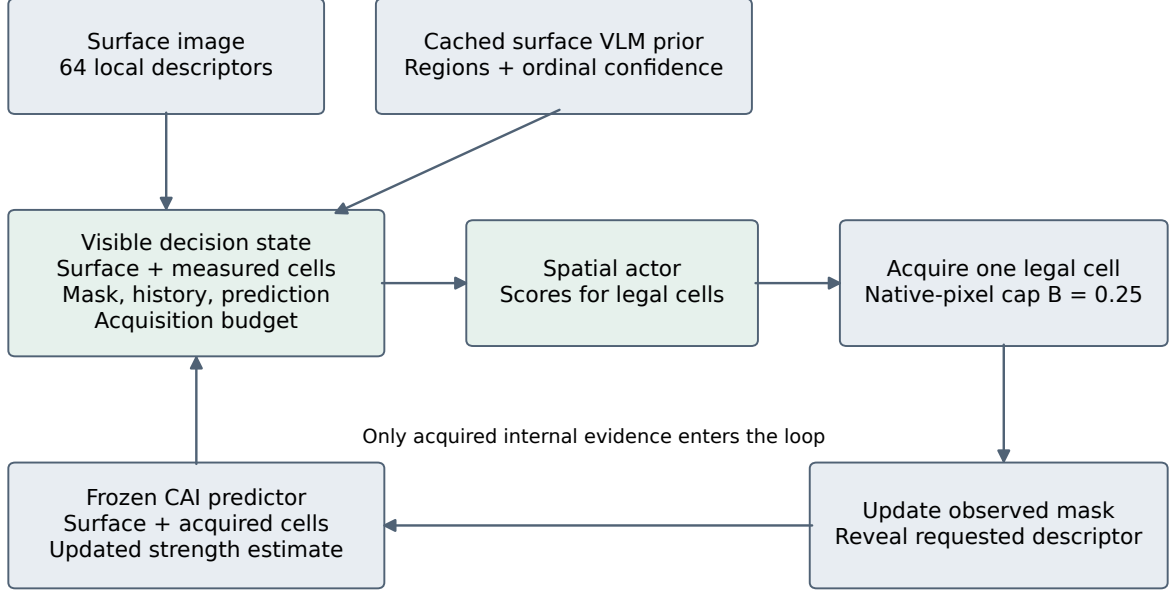
$$A(B) = \frac{1}{B} \int_0^B e(c) dc, \quad J = A(B) + 0.25e_T.$$

Both terms are measured in MPa and lower values are preferable. $A(B)$ measures prediction quality over the available acquisition range, whereas the terminal term retains an explicit incentive for final quality. The last prediction is held from c_T to B if no remaining whole cell is affordable. Consequently, a trajectory cannot avoid its final holding cost by ending below the cap. This formulation couples what is acquired with when its effect becomes available.

3.2 Multimodal inspection-state representation

A consistent spatial representation connects surface cues to candidate internal observations. Frozen ResNet18 encoders [9] produce a 512-dimensional descriptor for each of the 64 cells in each modality. Each cell is cropped before encoding, rather than extracted from a whole-image feature map that may already mix information across internal regions. The resulting surface and internal feature arrays each have shape 64×512 . Corresponding grid coordinates are normalized to $[0,1]$ in the row and column directions. These coordinates provide an explicit location reference for the predictor and acquisition policy; they do not establish a mechanically calibrated three-dimensional damage map.

Offline replay stores the complete internal feature array on the environment side so that any legal requested cell can be revealed. Model visibility is narrower than storage availability. Before cell



True strength: training and retrospective scoring only. Parameters stay frozen during acquisition.

Figure 1: Multimodal acquisition workflow. Only requested internal descriptors enter the visible state. The frozen VLM supplies a cached surface prior; the actor makes each acquisition decision. OpenAI Codex assisted in writing the deterministic plotting code.

mixing or spatial interaction, the predictor replaces unknown internal content with zero, and the spatial feedback actor replaces it with a learned mask vector. The binary observed flag distinguishes unknown cells from measured cells. Surface descriptors remain available everywhere. The actor also receives acquisition history and remaining budget, enabling it to distinguish a region that has not been visited from one that has already contributed to the current estimate.

A frozen Qwen2.5-VL-7B-Instruct model [10] supplies an additional surface prior. It examines clean and gridded surface views and returns candidate regions, ordinal confidence and cue-availability information. The acquisition actor receives numerical region indicators and confidence features, not the full generated explanation. Confidence is mapped from unknown, low, medium and high to 0, 1/3, 2/3 and 1, respectively; overlapping regions retain the highest confidence for a cell. These values encode an ordering of reported confidence rather than calibrated probabilities of internal damage. The VLM does not observe the hidden internal image, estimate CAI strength or produce a fresh language plan at every acquisition step.

The prior has a specific initialization role. At the first action, cells in the highest available reliable confidence level, restricted to medium or high, form a candidate set C_0 . The policy scores the affordable legal cells within that set. If the VLM is unavailable, reports no reliable cue, supplies no reliable region or leaves no affordable candidate, the ordinary legal set is retained. After the first acquisition, this hard restriction is removed while the numerical VLM features remain in the actor state. This design makes the initial prior explicit without constraining the entire sequence to a surface-selected region. Unavailability and an available response reporting no reliable cue are represented separately.

The information boundary can be summarized as follows.

Information	Acquisition actor during evaluation	Frozen CAI predictor	Training/scoring side
Complete surface descriptors	Available	Available	Available
Surface VLM region features	Available in VLM variants	Not an input	Available
Acquired internal descriptors	Available in feedback variants	Available	Available
Unacquired internal descriptors	Hidden	Hidden	Stored only by replay environment
Observed mask and acquisition fraction	Available	Available	Available
Action history and remaining budget	Available to state-dependent policies	Not an input	Available
Current predicted CAI strength	Available in feedback variants	Output	Available
True CAI strength	Excluded	Excluded from input	Supervision and scoring only

3.3 Partial-observation CAI assessment

A common assessment model is necessary to distinguish acquisition choices from changes in the downstream regressor. All nine strategies are evaluated with the same selected MEAN_SC predictor. Surface and masked internal descriptors are separately projected to 64 dimensions. For each cell, their concatenation with its observed indicator and two coordinates passes through a two-layer cell multilayer perceptron with GELU activations. This yields a 64-dimensional cell representation that contains local surface information and, where measured, internal information.

The predictor forms two summaries: the mean over all 64 fused cell representations and the mean over measured cells. The measured mean uses a denominator bounded below by one, which makes the initial empty state well defined. The all-cell mean is not a surface-only branch: measured internal content is already present in its constituent cell representations. Concatenating both summaries with the actual acquisition fraction gives the regression-head input. The normalized output is transformed to MPa using the training-derived target mean and scale. The same computation accepts an empty internal set, a partially observed set or all 64 cells.

Neither the policy identifier nor the acquisition order enters this predictor. Therefore, for the same surface, observed cells and cost, its output is identical regardless of the policy that produced the state. Different acquisition sequences can still have different $A(B)$, even when they eventually reveal the same subset, because intermediate predictions are available at different costs. This property gives the trajectory comparison its interpretation: changes in prediction quality arise through supplied observations and their timing, under a fixed assessment mapping.

Predictor training covers both sparse and dense observation sets. A random mask draw selects zero cells with probability 0.10, 1–16 cells with probability 0.60, 17–48 cells with probability 0.20, and all 64 cells with probability 0.10. Small sets use prefixes of Random, Center-first, Geometry-spread or Serpentine routes; larger partial sets use random permutations. These are sampling probabilities rather than fixed batch proportions. Specimens are sampled uniformly by domain and then within domain. Each batch contains 32 specimens and minimizes mean Huber loss on $(\hat{y} - y)/s$, with delta one and $s = \max(\text{std}(y_{\text{fit}}), 1 \text{ MPa})$. The same fit-derived mean and scale map the regression head to MPa. This strength supervision differs from the actor’s trajectory cost. AdamW uses learning rate 0.0003, weight decay 0.0001 and gradient-norm clipping at one. Checkpoints are evaluated every 250 updates, with at most 2000 updates and patience four. The candidate comparison evaluates fixed validation prefixes independently of actor trajectories. It selected MEAN_SC at update 1750 as P_{all} , rather than assigning it in advance. Dense training masks provide predictor input coverage

without extending the actor’s 0.25 acquisition budget.

During policy learning, a specimen in training fold k receives feedback from a frozen predictor P_{-k} fitted with its capture group excluded. Three such models use the selected architecture and fold-specific fit means and scales. Their checkpoints are still selected using the common validation library, so cross-fitting removes group overlap in fitting rather than all selection dependence. Evaluation in Algorithm 1 and the complete-input reference instead use P_{all} alone. The out-of-fold predictors are not averaged into a deployment ensemble.

3.4 State-dependent acquisition policy

The spatial actor represents each candidate location jointly with the current assessment state. Surface and visible internal descriptors are each projected to 48 dimensions. Their concatenation with row and column coordinates, the observed flag, normalized action history, VLM region indicator and confidence is mapped to a 128-dimensional cell token. A separate query token represents cumulative cost, remaining budget, normalized current CAI prediction, VLM availability, the no-reliable-cue indicator and the fraction of observed cells. These 65 tokens pass through two Transformer encoder layers [11] with four attention heads and a feed-forward width of 256.

The resulting query representation supplies global context to every cell. An action head scores each concatenated contextual cell and query pair; illegal cells are excluded before selection. A separate value head estimates the remaining training cost from the query representation. The value head is a critic used to reduce policy-gradient variation, not the predictor of reported CAI strength. The architecture thus connects a local choice to the current observation set while retaining a distinct assessment model. Its spatial interactions operate on already masked features, so the global context cannot retrieve hidden internal content through attention.

Policy-gradient learning [8] minimizes the expected trajectory objective over training specimens and sampled actions. Specimens are sampled by first choosing a domain uniformly and then a physical specimen within that domain. For each trajectory, the frozen cross-fitted predictor supplies absolute errors used to construct cost-to-go targets. The actor loss sums log action probabilities weighted by detached cost-to-go minus the critic estimate; a squared-error critic loss with weight 0.5 and an entropy term complete the optimization objective. Costs are normalized by the existing training target scale, and the discount factor is one. The shared static policy uses a zero baseline. This is task-loss-based policy learning, without an action-demonstration target.

During training, actions are sampled from the legal categorical distribution. During evaluation, the highest-scoring legal action is selected and the actor is evaluated again after the observation update. Parameters remain frozen during this sequence. The open-loop diagnostic removes internal descriptors and the current prediction from the decision computation but retains surface features, location, prior, acquisition history and budget. Its downstream predictor still incorporates the acquired internal information. The no-VLM diagnostic retains surface and internal feedback while removing the VLM features and initial restriction. These distinctions separate the information used to choose a cell from that used to assess the specimen afterwards.

Algorithm 1. Budget-limited task-driven acquisition

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Input: surface descriptors S; cached VLM features V;
      frozen actor pi_theta; evaluation predictor P_all;
      replay environment E; native cell costs d; total budget B=0.25
Initialize M=0, H=0, X_obs=0, c=0, t=0
p = P_all(S, X_obs, M, c)
while True:
    L = {i in {0,...,63}: M[i]=0 and c+d[i] <= B}
    if L is empty: break
    C = L
    if t=0 and highest-reliable VLM candidates overlap L:
        C = L intersect C0
    z = (S, X_obs, M, H, V, p, c, B-c)
    scores, _ = pi_theta(z)
    a = argmax of scores over C
    x_a = E.acquire(a)
    X_obs[a] = x_a
    M[a] = 1
    t = t+1
    H[a] = t/64
    c = sum of native cell costs over measured cells
    p = P_all(S, X_obs, M, c)
    Record action, observation state, cost and prediction
Return acquisition sequence and predictions
Compute errors with y only on the training/scoring side

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The algorithm exposes a logical observation buffer. The replay implementation stores the full descriptor array in the environment but masks unobserved content before model mixing, giving the same visible-information interface. The actor substitutes its learned unknown-cell vector wherever M is zero. At evaluation, each observation updates the prediction and decision state while both networks’ weights remain fixed. The value-head output is unused for action selection. Native-cost comparisons use the floating-point tolerance specified in Supplementary Section S1.

3.5 Acquisition timing and task cost

The trajectory objective admits a direct decomposition using saved prediction changes. For an acquisition completed at c_t , define $\delta_t = e_{t-1} - e_t$ and

$$g_t = (1 - c_t/B) \delta_t, \quad A(B) = e_0 - \sum_{t=1}^T g_t.$$

An error reduction contributes in proportion to the budget interval over which its updated prediction can influence the trajectory area. An error increase produces a negative contribution and is retained. An observation completed exactly at B has zero contribution to $A(B)$, although its prediction change can affect the terminal term in J. The identity also includes the holding interval after a trajectory ends below B; no additional extrapolated observations are required. Supplementary material gives the telescoping derivation.

For descriptive analysis, we group these signed contributions by acquisition completion stage: $(0,0.0625]$, $(0.0625,0.125]$, $(0.125,0.1875]$ and $(0.1875,0.25]$. A contribution assigned to a stage measures its weighted effect over the remaining budget, rather than an improvement confined to that stage’s integration interval. Aggregating first within an episode, then across repeated runs of each specimen, within domains and equally across domains preserves the estimand used for A. This is an accounting identity for the observed loss, not a causal allocation of gains to components or a new intervention on the acquisition policy.

4 Experimental design

4.1 Data and specimen correspondence

The study uses source image datasets [1] and CAI measurements [2, 12] assembled into an existing image-based composite-impact cohort comprising 276 physical specimens in six source domains. Each specimen is associated with a surface image, a registered internal C-scan crop and a CAI-strength measurement. Labels originate from the source authors’ workbooks, with measurement semantics explicitly identified as CAI strength and normalized to MPa. The specimen identifier, source version, worksheet and cell location are retained in the provenance records. Surface-to-internal correspondence is inherited from the registered specimen manifest, which records exact specimen identity and image hashes. The experiment replays these image regions; it does not collect new ultrasonic measurements or treat a displayed C-scan raster as a raw acoustic waveform.

Source relationships are handled at the capture-group level. Specimens sharing a source image identity, path or registered crop identity are joined transitively before splitting. The resulting 259 groups include 17 groups containing multiple physical specimens. A deterministic per-domain ordering assigns the first floor(0.6 n) groups to TRAIN, the next floor(0.2 n) to VALID and the remainder to TEST. All members of a group stay in the same partition. This produces 161 training specimens in 152 groups, 50 validation specimens in 48 groups and 65 reserved specimens in 59 groups. Table 1 lists physical counts by source domain; the supplementary provenance table supplies source versions and verified configuration descriptions.

Source domain and configuration	TRAIN	VALID	Reserved TEST	Total
74t7kcdgkr: 8-layer cross-ply [13]	25	9	11	45
cgtnjyggtm: 24-layer quasi-isotropic [14]	29	9	11	49
w68dtmpfyf: 16-layer quasi-isotropic [15]	25	8	10	43
xcmzfsbd9t: 24-layer cross-ply [16]	35	9	15	59
yfxyg8jm46: 16-layer cross-ply [17]	25	8	9	42
ykhs7s2dck: 8-layer quasi-isotropic [18]	22	7	9	38
Total physical specimens	161	50	65	276
Total capture groups	152	48	59	259

Table 1. Cohort composition. Configuration labels follow the version-1 source dataset descriptions. Per-specimen thickness is not inferred from layer count. All reported performance estimates use the 50 VALID specimens. Reserved TEST outcomes were not evaluated in this study.

The reported validation specimens were also used for checkpoint selection, and the underlying cohort includes specimens reused in earlier development. Accordingly, the results characterize selected validation performance rather than independent confirmation. Capture-group isolation addresses recorded image-source overlap across partitions, but does not by itself make this historically reused cohort an untouched external benchmark. No performance statement in this manuscript uses the 276-specimen total as its evaluation sample size.

4.2 Compared acquisition strategies

We compare nine acquisition strategies under the same legal-cell rule, native-pixel cost and frozen predictor. Four require no policy training. Center-first ranks cells by distance to the grid centre. Geometry-spread starts at a corner and repeatedly selects the cell with the greatest minimum squared grid distance from previously selected cells, with deterministic index-based tie breaking. Serpentine alternates horizontal direction across successive rows. Random uses a seeded permutation of all cells; five saved repeats are included. These are explicit acquisition rules for the replay experiment, rather than representations of all industrial inspection practices.

Five learned strategies test different levels of decision information. Learned-static has 64 shared position logits and receives no specimen image or prediction. VLM spatial open-loop uses the surface, prior and acquisition bookkeeping but excludes acquired internal content and prediction feedback from action choice. The main VLM spatial feedback policy adds those feedback channels. No-VLM spatial feedback removes the VLM channels and initial candidate restriction while retaining surface descriptors. VLM mean feedback retains the main information sources but replaces Transformer interactions with a mean-feedback architecture. The last comparison is not capacity matched: the spatial actors have 378,978 parameters, compared with 91,650 for the mean actor and 64 for learned-static.

Policy	Decision information	Structure	First-step VLM restriction
Center-first	Position only	Centre-distance order	No
Geometry-spread	Position only	Farthest-point order	No
Serpentine	Position only	Alternating row order	No
Random	Position only	Five seeded permutations	No
Learned-static	Shared position logits	64 parameters	No
VLM spatial open-loop	Surface, VLM, mask/history, cost	Two-layer spatial actor	Yes
VLM spatial feedback	Open-loop inputs plus internal/prediction feedback	Two-layer spatial actor	Yes
No-VLM spatial feedback	Surface, internal/prediction feedback, bookkeeping	Two-layer spatial actor	No
VLM mean feedback	Same information as main	Mean-feedback actor	Yes

Table 2. Acquisition-policy information and structure. The state-permission table in Section 3 specifies channel visibility. Across all controls, the assessment model still receives every actually acquired internal cell. The complete-input reference is listed separately at fraction 1.0 and uses all 64 internal cells plus surface descriptors with the same predictor. It has no observed acquisition trajectory and therefore no A(B).

4.3 Learning and evaluation protocol

Perception and assessment were frozen before the evaluated policy runs. The fit-side VLM cache contains 211 specimen records, of which 205 are available and six are terminally unavailable. The latter remain distinct from valid no-cue responses. Three common-predictor candidates were trained: MEAN_SC, a spatial surface-plus-internal predictor and a spatial internal-only predictor. All passed the predefined prediction-quality eligibility checks. Selection minimized area over the fixed validation-prefix library, averaging four routes within specimen, then within domain and equally across domains. Ties within 10^{-8} were resolved by parameter count and then model name. This procedure selected MEAN_SC at update 1750; the three cross-fitted training-feedback predictors were selected at updates 1250, 750 and 1000. Policy evaluation uses the common predictor only. Predictor-specific settings and eligibility checks are listed separately from actor settings in Supplementary Section S1.

Policies were optimized with AdamW [19], learning rate 3×10^{-4} , weight decay 10^{-4} , batch size 16 and gradient-norm clipping at one. Entropy regularization decreased linearly from 0.01 to zero over each prescribed run. The value-loss weight was 0.5, the discount factor was one and the terminal-error weight was 0.25. Every 250 updates, deterministic validation trajectories were scored by six-domain-equal A(B). The selected checkpoint minimized this measure, with an improvement tolerance of 10^{-12} and patience of four validation evaluations. Four policies each ran for 1250 updates and learned-static for 750, giving 5750 actual updates and 23 saved candidate evaluations.

One initialization was used per learned method, with seeds 2026091301–2026091305 assigned to main, no-VLM, open-loop, static and mean policies, respectively. Selected updates were 250, 1000, 750, 250 and 250 in that order. Random repeats used seeds 2026091250–2026091254. These method-specific initializations form one policy-seed panel, not repeated training seeds for each method. Evaluation comprises 650 saved method–specimen–repeat episodes, containing 10,227 acquisition events. Runs and events are dependent descriptions of 50 physical specimens and are not additional independent observations. Detailed training settings and software versions are retained in the supplementary material.

4.4 Cost, quality and statistical analysis

The main common-cap grid is $\{0, 0.0625, 0.125, 0.1875, 0.25\}$. All lower-cap estimates are queried from prefixes of trajectories generated with a total acquisition budget of 0.25; the policy is neither retrained nor rerun with a different total budget for each reported cap. The actor therefore receives remaining budget $0.25 - c_t$, rather than a remaining budget defined by the queried cap. At each cap, we use the last prediction whose acquisition cost is at or below that cap. Actual fractions are reported separately because complete native cells may leave an unspent remainder. A supplementary event grid contains the 599 shared breakpoints obtained from saved episodes. Both grids retain current-state prediction reversals, with no monotonic correction or extrapolation beyond 0.25. Full-input quality is supported only at fraction 1.0.

At each cap, absolute and squared losses are averaged across Random repeats within each physical specimen and then pooled over the 50 specimens. RMSE is the square root of pooled MSE. R^2 is computed across specimens within each repeat and then averaged; predictions are not ensembled. In contrast, A and early A first average episode losses within a specimen, then average specimens within each domain, and finally weight the six domains equally. Early A uses an integration endpoint of 0.0625. These different aggregations mean a pooled endpoint metric and a domain-equal trajectory metric need not rank policies identically.

Reported uncertainty comes from the existing 5000-replicate, within-domain capture-group bootstrap, using seed 2026091401. Resampled groups carry all their physical members with their sampled multiplicities; the same saved weights support paired method comparisons. The 95% intervals are pointwise exploratory intervals conditional on selected validation checkpoints. They do not correct checkpoint selection, repeated historical use, post hoc analysis choices or multiple comparisons. We report their numerical ranges rather than interpreting them as independent confirmatory tests.

Matched-quality analysis first computes each method’s cohort-level MAE curve and then finds its earliest supported cap $b^*(q)$ satisfying $\text{MAE}(b) \leq q$. Relative acquisition reduction against a control is $1 - b_{\text{main}}^*(q)/b_{\text{control}}^*(q)$, when both are reached and the control denominator is positive. The integer targets 41–61 MPa and 21 source-labelled anchors, including full-input quality, are descriptive empirical targets rather than prespecified engineering acceptance limits. Main-grid and event-grid results remain separate. Unreached targets, zero denominators, negative reductions and later recrossings are retained. The calculation compares population curves; it does not choose a label-informed stopping time for individual specimens.

5 Results and discussion

5.1 Prediction quality under matched acquisition caps

Task-driven feedback produced lower observed errors than several fixed acquisition rules under the common 25% cap. The main policy achieved a pooled MAE of 44.286 MPa, RMSE of 58.517 MPa and R^2 of 0.659 on the 50 validation specimens. Geometry-spread achieved 46.910 MPa MAE, giving a paired improvement of 2.624 MPa with a 95% exploratory interval of $[-0.488, 5.925]$ MPa. The actual acquired fraction for the main policy averaged 0.247437 and ranged from 0.234492 to

0.249999. The comparison is therefore at a matched cap, with small specimen-dependent differences in the completed native-pixel fractions.

Across the entire acquisition range, the main policy had a six-domain-equal A of 45.110 MPa, compared with 47.311 MPa for geometry-spread. The lower area indicates that its advantage was not confined to the terminal estimate. However, area and terminal quality capture different properties: Random had a slightly lower endpoint MAE than geometry-spread but a higher A. Inspecting only the final state would miss this difference in the quality supplied during acquisition. Figure 2 retains the observed stepwise MAE paths, including reversals, and Supplementary Table S2 reports all five common caps with their actual acquisition ranges.

Acquisition policy	A (MPa)	Early A (MPa)	MAE at cap (MPa)	RMSE at cap (MPa)	R ² at cap
Center-first	48.475	51.821	48.027	65.223	0.577
Geometry-spread	47.311	51.880	46.910	62.278	0.614
Serpentine	48.911	52.170	48.013	63.041	0.605
Random	49.594	54.117	46.886	61.941	0.618
Learned-static	47.977	52.946	46.993	62.168	0.616
No-VLM spatial feedback	43.597	48.771	42.385	56.534	0.682
VLM mean feedback	45.263	50.390	43.611	59.012	0.654
VLM spatial feedback (main)	45.110	49.642	44.286	58.517	0.659
VLM spatial open-loop	46.501	51.048	44.822	59.663	0.646
Complete input, fraction 1.0	—	—	41.690	53.944	0.711

Table 3. Common-predictor comparison. Partial policies use cap 0.25; the complete-input row is a separate cost-1.0 reference. A integrates over 0–0.25 and early A over 0–0.0625, each normalized by its own interval. Areas weight domains equally; endpoint measures pool physical specimens after repeat-loss averaging. All values are selected-validation estimates, with n=50 physical specimens. Actual fraction ranges and full paired results are provided in the supplementary source tables.

5.2 From a shared order to evidence-dependent acquisition

The three policies spanning a shared ranking, specimen-conditioned open-loop choice and internal feedback had progressively lower area estimates: 47.977, 46.501 and 45.110 MPa. This sequence connects the experiment to the intended engineering distinction. Learned-static offers one order for every specimen. Open-loop can use surface variation to allocate observations differently, while feedback can revise its allocation after seeing internal content. Their performance pattern is consistent with useful specimen dependence, followed by further improvement when current acquired evidence participates in the next decision.

The main policy improved on open-loop by 1.391 MPa in A and 0.536 MPa in endpoint MAE. The area direction favoured feedback in five of the six domains. The comparison therefore links feedback to a lower observed trajectory loss over much of this cohort, while the interval in Table 4 retains uncertainty about the difference. These adjacent contrasts do not identify percentages of total improvement caused by individual modules: policies were separately trained, used different initializations and visited different states. Their value is to show the consequences of distinct decision-information arrangements under the same assessment model.

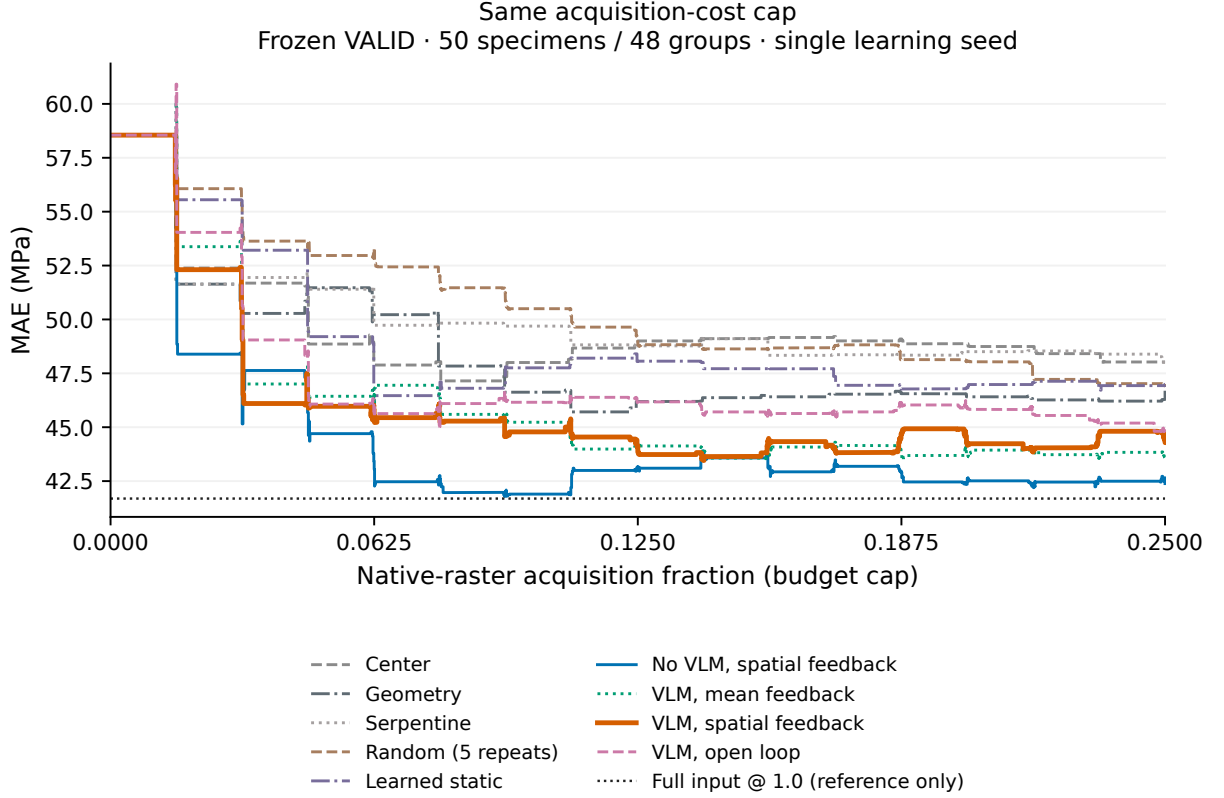


Figure 2: MAE along the recorded acquisition range. Curves are held current-state values on the shared event grid, with no smoothing. The complete-input horizontal reference represents quality measured only at cost 1.0.

Contrast, control minus main	Measure	Difference (MPa)	95% exploratory interval (MPa)
Geometry-spread versus main	A	2.200	[−0.787,5.033]
Open-loop versus feedback	A	1.391	[−0.683,3.406]
No-VLM versus VLM	Early A	−0.872	[−4.870,3.255]
Mean versus spatial feedback	A	0.152	[−2.186,2.425]

Table 4. Component-related comparisons. Positive values favour the main policy for the stated measure. All four intervals contain zero. Intervals use the existing paired within-domain capture-group bootstrap and are conditional on selected validation checkpoints. Supplementary Figure S4 displays these contrasts; the VLM contrast uses early A rather than full-range A.

The no-VLM feedback policy obtained lower full-range A and endpoint MAE than the main policy, at 43.597 and 42.385 MPa, respectively. Its surface descriptors were still present, so this result concerns the additional VLM prior rather than the utility of surface information as a whole. Spatial interaction likewise provided a mixed comparison with mean feedback: the main actor’s area was lower by 0.152 MPa and its RMSE by 0.495 MPa, but its endpoint MAE was higher by 0.675 MPa. These results support keeping the components conceptually distinct instead of treating architectural complexity as evidence of a uniform performance gain.

5.3 When acquired evidence contributes to prediction quality

The largest positive timing difference between feedback and open-loop acquisition occurred after the initial internal observations had become available. Main-minus-open-loop contributions across the four completion stages were +0.415, +1.296, -0.223 and -0.097 MPa. Their net sum corresponds to the main policy’s 1.391 MPa lower error area. The second stage, spanning completion fractions above 0.0625 and up to 0.125, provided the largest positive difference. Thus, the observed trajectory difference cannot be summarized solely as selecting a better starting region.

The common predictor gives this comparison a specific interpretation. Each acquired subset changes the evidence supplied to the same regression mapping, and the completion time determines how long that updated estimate enters the area objective. Contributions assigned to the second stage are weighted over the remaining interval to B; the 1.296 MPa difference is not a local integral reduction confined to that stage. The main policy’s absolute stage contributions were 11.031, 0.960, -0.091 and 0.038 MPa. Subtracting their sum from the common domain-equal initial error of 57.049 MPa gives A=45.110 MPa. This initial error differs from the pooled initial MAE of 58.551 MPa because the two estimands weight domains differently.

The timing pattern also distinguishes feedback from broad spatial coverage and a learned shared order. Geometry-spread obtained first- and second-stage contributions of 7.143 and 2.672 MPa; its smaller first-stage value was partly offset in the second stage. Learned-static obtained 9.729 MPa in the first stage, followed by -1.058 MPa in the second. Against open-loop, the main policy’s second-stage contribution was positive while the control’s was negative. This pattern is consistent with useful evidence-dependent allocation once initial internal information is available. The separately trained policies nevertheless differ in initialization and visited states, so the decomposition accounts for observed prediction changes rather than assigning a causal share to feedback.

Later observations qualified the earlier improvement. The final two main-versus-open-loop differences were negative, and 381 of the main policy’s 793 acquisitions increased individual absolute error. Revealing a descriptor can move a finite regressor’s estimate away from the measured strength, even within a sequence with lower overall area. Retaining these signed changes makes the accounting compatible with the reversals in the quality curves. Useful state-dependent acquisition therefore concerns the resulting evidence sequence, rather than guaranteed error reduction after every action.

Figure 4 follows the existing c8-16 case after one and eight acquisitions, showing how the measured set and actual next action change as evidence accumulates. The first proposal was narrower than the ordinary legal set for 45 of 50 main-policy specimens; subsequent decisions used the expanded legal set and updated state. These panels illustrate that progression without changing the original observed content. The same three fixed cases remain in the supplement, including the higher-error q24-48 case and their prediction trajectories. Case images document executed choices; the timing comparison above summarizes their cohort-level prediction consequences.

5.4 Acquisition requirements at matched empirical quality

An acquisition advantage depends on the target quality and the comparator. At $q=46.909917$ MPa, the geometry-spread endpoint MAE used as one of the complete anchor set, the main policy first met the target at cap 0.0625 on the five-point grid. Geometry-spread first met it at 0.125, giving a 50% reduction in the earliest supported cap. Learned-static also first met it at 0.0625, giving zero reduction relative to that comparator. Using geometry-spread’s nominal 0.25 endpoint as its required cost would overstate the reduction, because that method had already reached the same empirical quality earlier.

The finer event grid resolves earlier crossings but also reveals their instability. For the same target, the earliest main, geometry and static caps were 0.031388, 0.093426 and 0.062407, respectively. These correspond to reductions of 66.40% and 49.70% against geometry and static on that grid. At the main policy’s crossing, its actual mean fraction was 0.030861 and MAE was 46.507051 MPa;

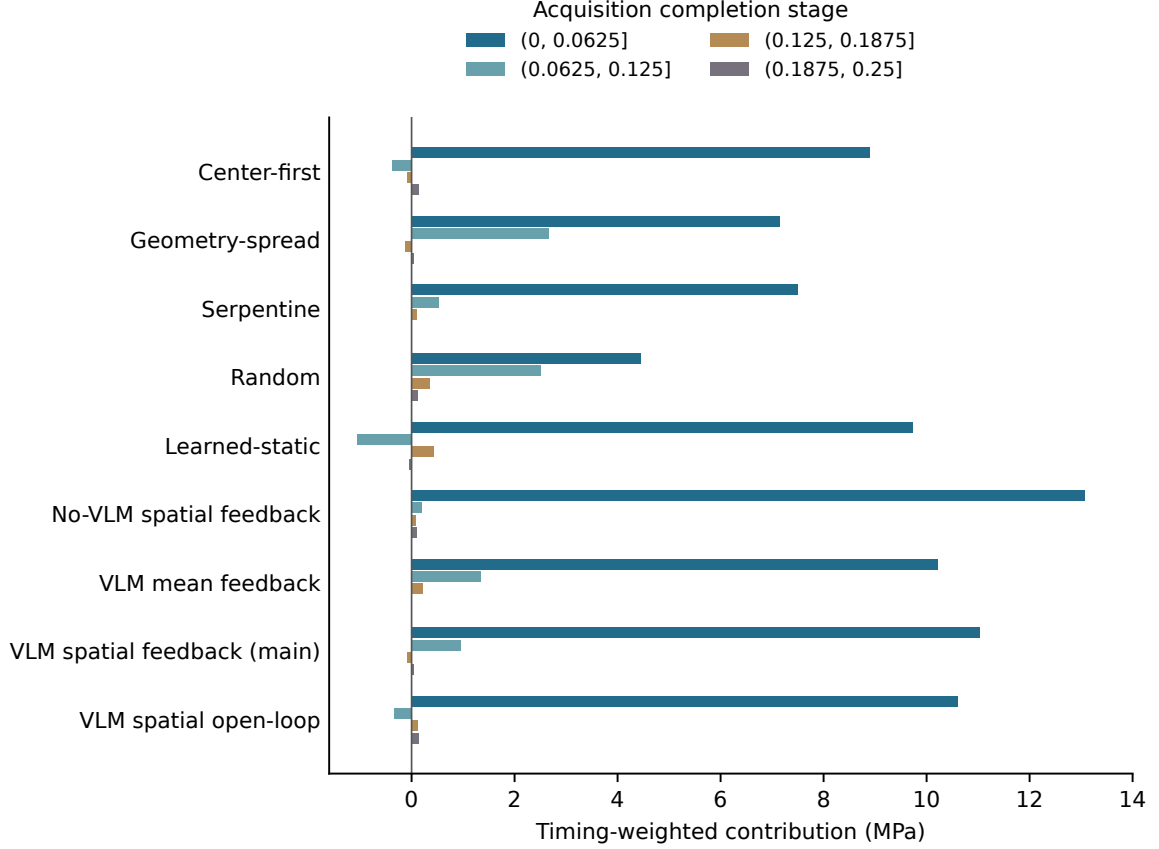


Figure 3: Timing-weighted error contributions grouped by acquisition completion stage. All nine methods and negative values are retained. Episode sums are averaged over repeats within specimen, within domain and equally across six domains, with 50 physical specimens. Contributions affect the remaining budget interval after completion; they are not stage-local error integrals.

its curve subsequently rose above the target. We retain this event-grid result in the supplementary analysis, separately from the primary 50% and 0% comparisons, because grid resolution and first-passage reversals materially change its interpretation.

Figure 5 presents the full main-grid target range rather than a single favourable threshold. Supplementary tables retain all integer targets and source-labelled anchors on both grids, including unreached targets and negative or undefined ratios. This representation answers a practical descriptive question: at what supported cap did each population curve first achieve a specified quality? It does not supply an individual stopping certificate. An operational stopping policy would need information available without knowing the specimen’s true CAI strength, and the present first-passage calculation does not provide that capability.

5.5 Full-information trade-off and engineering interpretation

The common predictor with complete internal and surface input achieved 41.690 MPa MAE, 53.944 MPa RMSE and $R^2=0.711$. The main policy at the 25% cap therefore retained a 2.596 MPa MAE gap to complete input. None of the nine partial-acquisition policies reached the complete-input MAE anywhere in either observed grid. The full-input point helps locate the quality cost of partial observation under this predictor, but does not define an optimal industrial reference or a missing trajectory between fractions 0.25 and 1.0.

The engineering contribution is the explicit coupling between an assessment objective and the orga-



Figure 4: Acquisition progression for the preselected c8-16 specimen after one and eight acquisitions, ordered left to right. Both panels preserve the original native-cost annotations, next-cell identifier and observed/unobserved colors; grey cells remain unobserved. Panels show acquisition steps, not a matched-cost performance comparison. Source imagery: Hasebe et al., version-1 dataset, CC BY 4.0; the registered overlays are assembled unchanged.

nization of acquired information. A policy can be compared at a shared cap, through its complete observed error trajectory, and at a shared empirical quality target. The timing contrast explains why area and terminal MAE need not agree: an observation changes the error over the remaining acquisition interval, not just at the endpoint. Together with target-dependent first passages, this connects sequential information acquisition to the image-based assessment objective. The contribution is a way to organize and evaluate observations according to that objective, rather than a new regression architecture or a universal acquisition-reduction percentage.

Domain-level results further bound the comparison. Against geometry-spread, the main policy’s area difference favoured it in three domains and favoured the control in three. Against open-loop, five domain differences favoured feedback. This heterogeneity is compatible with acquisition decisions depending on specimen evidence, but it also leaves room for differences in predictor quality, source imaging and the distribution of specimens across domains. The reported data do not determine a material-specific causal explanation. The supplementary domain table preserves these directions without treating the six domains as six independent replications of the entire learning experiment.

The current evidence supports an offline image-acquisition method and a quality–acquisition analysis on a selected validation cohort. Physical deployment remains a separate question because native image pixels omit probe travel, coupling, setup and measurement latency; a 25% image fraction cannot be translated directly into a 75% time saving. A single policy-seed panel and validation-based selection also limit inference about reproducibility and transport to new specimens. The results establish the observed quality–acquisition relationships within that replay protocol; their transfer to independent specimens and physical inspection operations remains to be evaluated.

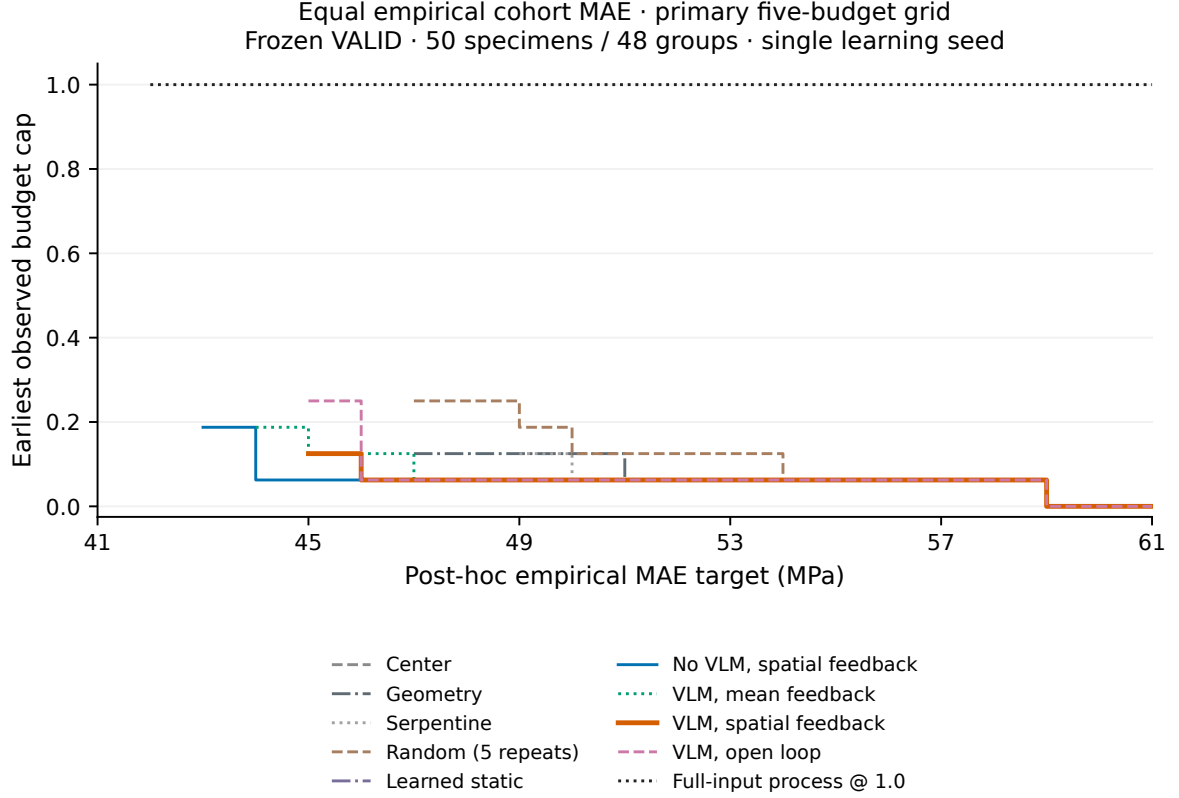


Figure 5: Earliest supported cap at each empirical MAE target on the primary five-cap grid. Missing points denote unreached quality; the finer event-grid results remain separate. The curves summarize cohort-level first passages, not a specimen stopping policy.

6 Conclusions

State-dependent acquisition connects the choice of internal observations to the evolving compression-after-impact assessment. The method combines surface priors, acquired C-scan descriptors and the current strength estimate to select subsequent cells. A separate frozen predictor keeps the assessment mapping common across strategies, while the learning objective values prediction quality throughout the acquisition budget and at its endpoint.

Offline replay on 50 validation specimens produced a main-policy MAE of 44.286 MPa under the 25% cap, compared with 46.910 MPa for geometry-spread. Relative to surface-conditioned open-loop acquisition, the largest positive timing contribution difference occurred in the second completion stage. This locates the observed advantage beyond initialization, while later negative differences show that improvement was not sustained at every stage. Component comparisons remained mixed, and partial acquisition did not reach the 41.690 MPa complete-input MAE.

The framework characterizes how acquired subsets and their availability over time shape CAI prediction quality. Same-cap comparisons, signed timing contributions and matched-quality requirements provide complementary views of that relationship. The results are conditional on validation-based checkpoint selection and one policy-seed panel; they describe image-acquisition trade-offs rather than independently validated physical inspection savings.

Declarations

Author details, contributions, funding, competing interests and final data/code release information: [to be completed by the authors; see AUTHOR_INPUTS.md]. Public source datasets and version-specific DOIs are cited in the manuscript.

During manuscript preparation, OpenAI Codex assisted with drafting, organization, English expression, reference screening and verification support, figure-programming assistance and document preparation. The exact underlying model identifier remains to be verified. Research use of Qwen2.5-VL is described separately in Section 3. Human scientific review and approval of the final manuscript remain pending; the authors retain responsibility for the submitted content.

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